

MARS SOCIETY 2026

Mars Harness Architecture Guide

A staged, evidence-controlled operating model for evolving mission processes.

THE INTELLIGENCE LAYER FOR MARS

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The operating premise

A model can describe how to perform a task; a harness determines what it can reliably do. Mission performance therefore depends on the system around the model: its loop, instructions, tools, context, memory, specialists, validation, and authority boundaries.

Seven functions

#	Harness function	Operational question
1	Orchestration loop	How does the agent plan, act, inspect, and iterate?
2	Instructions	What role, purpose, policies, and definition of done govern it?
3	Tool access	Which actions can it actually execute, and with what permissions?
4	Context management	What evidence enters the current decision window?
5	Memory + state	What durable facts, decisions, versions, and checkpoints persist?
6	Subagents	Which specialist roles can be delegated bounded work?
7	Guardrails + approvals	What is blocked, reviewed, escalated, or reversible?

Staged specialist sequence

Stage	Primary harness	Promotion condition
Discover	Hypatia / Hyperia	Decision-relevant evidence ledger
Engineer	Intelligent Engineer	Requirements, options, interfaces, verification plan
Verify	Ground-Truth Gatekeeper	Pass or conditional pass against explicit tests
Deliver	PM Accelerator / Vibe Engineering	Owned, reproducible release with rollback
Learn + Evolve	Mission Memory Steward	Outcome recorded; approved playbook change versioned

Decision loop

Frame → plan → execute with least privilege → inspect evidence → validate → approve → commit state → learn. Each stage consumes and emits a work-product contract. Failure returns exact remediation; it does not disappear into another prompt.

Risk architecture

Risk tier	Allowed autonomy	Required control
Low	Drafting, classification, reversible analysis	Automated checks and logged output
Medium	Tool use in bounded workspace	Independent validation and named owner
High	Safety, production, credentials, funds, physical systems	Human approval before action; rollback and audit trail

Evolution without process drift

Treat improvement as controlled configuration management: measure outcomes, identify failure modes, propose a bounded change, test it on replay cases, review the evidence, version the instruction or tool map, and retain the superseded procedure. The harness evolves because outcomes are incorporated—not because memory silently rewrites the rules.